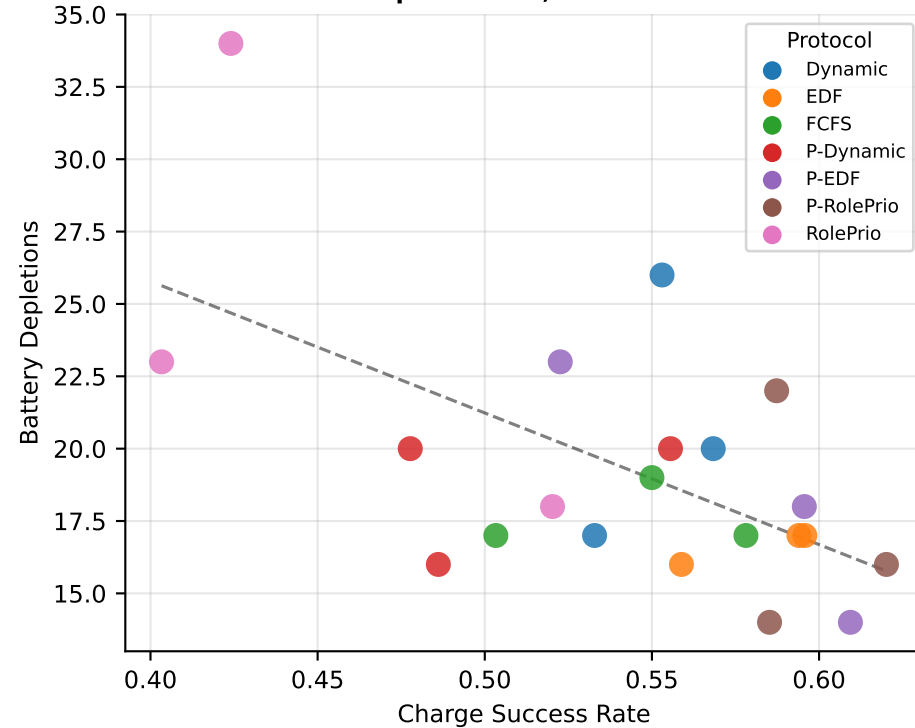


Cross-Layer Mechanism: Charging → Survivability → Network (Three-step causal chain; n = 21 runs)

Charging → Fleet Survival

$r = -0.58$, $\rho = -0.49$

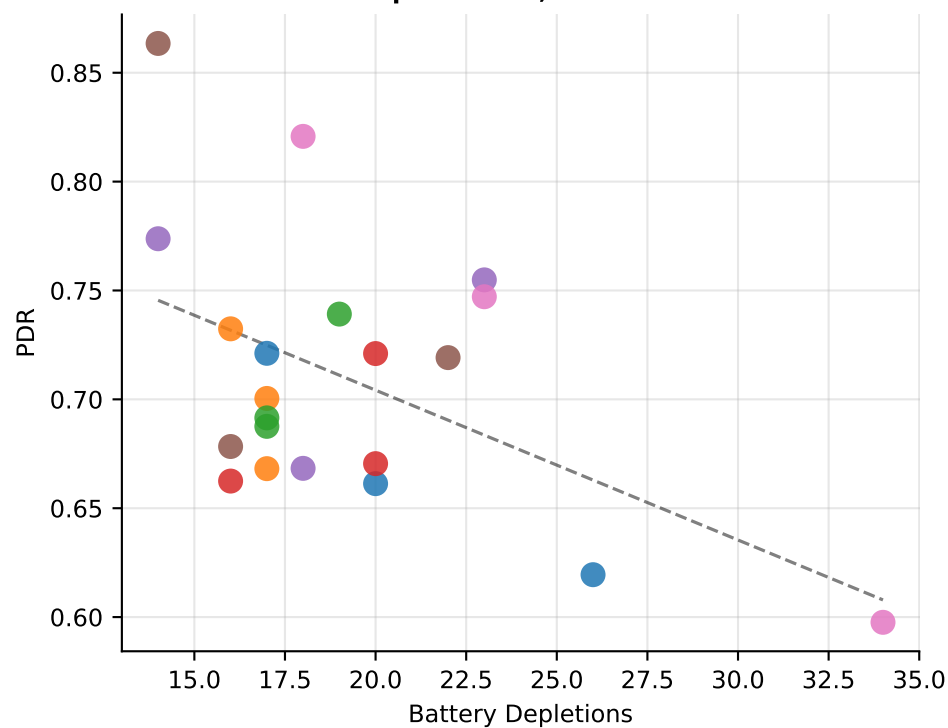
$p = 0.0055, 0.0229$



Fleet Survival → Network PDR

$r = -0.50$, $\rho = -0.28$

$p = 0.0197, 0.2121$



Routing Drops → Network PDR

$r = 0.22$, $\rho = 0.23$

$p = 0.3374, 0.3116$

